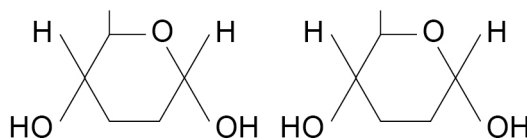


0 3

Figure 2 shows the structure of two α -glucose molecules.

Figure 2



0 3 . 1

On **Figure 2**, draw a box around one chemical group in each glucose molecule used to form a glycosidic bond.

[1 mark]

0 3 . 2

A precipitate is produced in a positive result for reducing sugar in a Benedict's test. A precipitate is solid matter suspended in solution.

A student carried out the Benedict's test. Suggest a method, other than using a colorimeter, that this student could use to measure the **quantity** of reducing sugar in a solution.

[2 marks]

[Extra space]

Question 3 continues on the next page

Turn over ►



In an investigation, a student wanted to identify the solutions in two beakers, **A** and **B**. She knew one beaker contained maltose solution and the other beaker contained glucose solution. Both solutions had the same concentration.

She did two separate biochemical tests on a sample from each beaker.

Test 1 – used Benedict’s solution to test for reducing sugar.

Test 2 – added the enzyme maltase, heated the mixture at 30 °C for 5 minutes, and then used Benedict’s solution to test for reducing sugar.

Maltose is hydrolysed by maltase.

The student’s results are shown in **Table 1**.

Table 1

	Colour of solution after testing with Benedict’s solution	
Beaker	Test 1	Test 2
A	red	red
B	red	dark red

0 3 . 3

Explain the results for beakers **A** and **B** in **Table 1**.

[2 marks]

Beaker A

Beaker B



0 3 . 4

Use of a colorimeter in this investigation would improve the repeatability of the student's results.

Give **one** reason why.

[1 mark]

In **Test 1**, the student used a measuring cylinder to measure 15 cm^3 of solution from a beaker. The measuring cylinder gives a volume with an uncertainty of $\pm 1 \text{ cm}^3$. She used a graduated syringe to measure 5.0 cm^3 of Benedict's solution. The graduated syringe gives a volume with an uncertainty of $\pm 0.5 \text{ cm}^3$. She mixed these volumes of liquid to do the biochemical test.

0 3 . 5

Calculate the percentage error for the measurements used to obtain a 20 cm^3 mixture of the solution from the beaker and Benedict's solution. Show your working.

[2 marks]

Answer = _____ %

8

Turn over for the next question

Turn over ►

